

DRAWING MACHINE

Finish off the code with a user interface

The drawing machine needs an interface to make it easier to use. This will let the user enter a string from the keyboard to tell the machine what to draw.

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This code creates a pop-up window where the user can input instructions. A “while True” loop lets them keep entering new strings.

The triple quote (”) tells Python that everything until the next triple quote is part of the same string, including the line breaks

```
instructions = '''Enter a program for the turtle:
```

```
eg F100-R45-U-F100-L45-D-F100-R90-B50
```

```
N = New drawing
```

```
U/D = Pen Up/Down
```

```
F100 = Forward 100
```

```
B50 = Backwards 50
```

```
R90 = Right turn 90 deg
```

```
L45 = Left turn 45 deg'''
```

```
screen = getscreen()
```

```
while True:
```

```
    t_program = screen.textinput('Drawing Machine', instructions)
```

```
    print(t_program)
```

```
    if t_program == None or t_program.upper() == 'END':
```

```
        break
```

```
    string_artist(t_program)
```

Tells the user what letters to use for different turtle commands

End of the string

Gets the data needed to create the pop-up window

This line tells the program what to show in the pop-up window

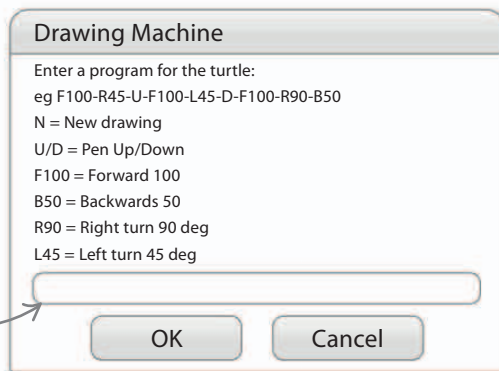
Stops the program if the user types “END” or presses the “Cancel” button

Passes the string to the String artist function

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This window pops up over the turtle window ready for the user to type a drawing machine program string.

Type the program string here and then click “OK” to run the program



△ Turtle control

Using this program, the turtle is easier to control, and you don't have to restart the program to draw another picture.